Uni-variate data

Uni-variate data is a vector of dependent variables, but since each data-point typically has an associated independent data point, uni-variate data is specified as set of pairs of values for an independent and associated dependent variable. Auxiliary data might be associated to any data, so also to uni-variate data.

Predictions are presented graphically as a curve by default, but sometimes interpolation of the independent variable is not possible. Alternatively, predictions can be presented as (open blue) markers, so interpolation is avoided, by specifying auxiliary data with the name `treat` for that data set, as a cell string with first element zero, like `auxData.treat.tW = {0};` for data set `tW`. The mydata-file for Bolinopsis mikado gives an example.

Codification of types uni-variate data

In the mydata file the author needs to specify which types of data are used in the entry. Labels for uni-variate data are packed as metaData.data_1 in the mydata-file. The codification generally follows DEB's notation rules. The table below presents the labels for the different types of univariate data found in the collection (dark yellow cells to the far right of each row in the species list). For the labels the independent and the dependent variables are separated by "-".

Optionally this can be followed by an underscore ("_"), indicating several values of a modifying factor:

f	functional response	T	temperature
L	length	W	weight

Several f or T might mean several constant values, but also that they vary in time. Various labels relate to life history events, as listed for zero-variate data. Trailing e means that data refers to embryo development. Data might relate to males and females; this is not indicated in the data labels, but comments in such entries specify the parameters in which they differ. Filtering rate also includes food-searching rate (for species that don't filter at all).

Data labels	Description
t-Le	time, embryo length
t-Le_T	time, embryo length (several T)
t-Le_fT	time, embryo length (several f, T)
t-L	time, length
t-L_T	time, length (several T)
t-L_f	time, length (several f)
t-L_fT	time, length (several f, T)
t-LR	time, length including gonads
t-dL	time, change in length
t-Ae	time, surface area for embryo
t-A	time, surface area
t-V	time, volume
t-V_f	time, volume (several f)
t-VYe	time, embryo yolk volume
t-VY	time, yolk volume
t-VY_T	time, yolk volume (several T)
t-Wwe	time, embryo wet weight
t-WwYe	time, embryo yolk wet weight
t-WwVe	time, embryo wet weight excluding yolk
t-Ww	time, wet weight
t-Ww_f	time, wet weight (several f)
t-Ww_T	time, wet weight (several T)
t-Ww_fT	time, wet weight (several f, T)
t-WwR	time, gonad wet weight
t-WdVe	time, embryo dry weight (excluding yolk)
t-WdYe	time, embryo dry weight (yolk)
t-Wde	time, embryo dry weight (total)
t-Wde_fT	time, embryo dry weight (several f, T)
t-Wd	time, dry weight
t-Wd_f	time, dry weight (several f)
t-Wd_T	time, dry weight (several T)
t-Wd_fT	time, dry weight (several f, T)
t-WdR	time, gonad dry weight
t-WC	time, carbon weight
t-WC_T	time, carbon weight (several T)

t-M_PLC	time, mass of protein, lipid, carbohydrate
t-M_N	time, mass of nitrogen
t-M_DNA	time, mass of DNA
t-M_RNA	time, mass of RNA
t-E	time, energy content
t-pX	time, food energy ingestion rate
t-p+	time, heat production
t-R_f	time, reproduction rate (several f)
t-R_T	time, reproduction rate (several T)
t-N	time, number of offspring
t-N_f	time, number of offspring (several f)
t-N_T	time, number of offspring (several T)
t-N_fT	time, number of offspring (several f, T)
t-JX_f	time, ingestion rate (several f)
t-F_f	time, filtering rate (several f)
t-JX	time, ingestion rate
t-JX_T	time, ingestion rate (several T)
t-JCe	time, embryo CO_2 production
t-JOe	time, embryo O_2 consumption
t-JOe_T	time, embryo O_2 cons (several T)
t-JO	time, O_2 consumption
t-JNe	time, embryo NH_3 production
t-T	time, body temperature
t-S	time, surviving fraction
t-S_f	time, surviving fraction (several f)
t-S_T	time, surviving fraction (several T)
t-S_fT	time, surviving fraction (several f, T)
t-dC	time, condition factor
ax-Wwx_f	age, weight at fledging (several f)
X-dL_L	food dens., change in length (several L)
X-dWd_W	food dens., change in dry w. (several W)
X-aj	food density, age at j
X-ap	food density, age at p
X-am	food density, age at death
X-Lp	food density, length at p
X-Li	food dens., length at i
X-Vp	food density, volume at p
X-Vi	food density, volume at i
X-Ri	food density, reproduction rate at i
X-JX	food density, ingestion rate
X-JX_L	food density, ingestion rate (several L)
X-JX_W	food density, ingestion rate (several W)
X-JP_L	food density, feaces production (several L)
X-JC_W	food density, CO_2-production (several W)
L-L	length, length (different length measures)
L-dL	length, change in length
L0-Lt	length length, initial and after a known period
L-V	length, volume
L-Wwe	length, embryo wet weight
L-Ww	length, wet weight
L-Ww_f	length, wet weight (several f)
L-Wde	length, embryo dry weight
L-Wde_T	length, embryo dry weight (several T)
L-Wd	length, dry weight
L-Wd_f	length, dry weight (several f)
L-WwR	length, gonad wet weight
L-dWw	length, change in wet weight
L-WC	length, carbon weight
L-WN	length, nitrogen weight
L-E	length, total energy
L-N	length, number of eggs/offspring
L-N_f	length, number of eggs/offspring (several f)
L-N_T	length, number of eggs/offspring (several T)
L-R	length, reproduction rate
L-GSI	length, gonado somatic index
L-F	length, filtering rate
L-JX	length, food consumption
L-X	length, cumulated food ingested
L-JC	length, CO_2 production
L-JO	length, O_2 consumption
L-JO_T	length, O_2 consumption (several T)
V-V	volume at t_1, volume at t_2

V-V	volume at t_1, volume at t_2
Ww-dWw	wet weight, change in wet weight
Ww0-Wwt	wet weight at capture, wet weight at recapture
Ww-dWw_T	wet weight, change in wet weight (several T)
Ww-N	wet weight, number of eggs/offspring
Ww-N_T	wet weight, number of eggs/offspring (several T)
Ww-R	wet weight, reproduction rate
Ww-R_f	wet weight, reproduction rate (several f)
Ww-JX	wet weight, ingestion rate
Ww-JX_T	wet weight, ingestion rate (several T)
Ww-JX_f	wet weight, ingestion rate (several f)
Ww-X	wet weight, cumulated food ingested
Ww-JO	wet weight, O_2 consumption
Ww-JO_T	wet weight, O_2 consumption (several T)
Ww-JO_f	wet weight, O_2 consumption (several f)
Ww-JN_T	wet weight, NH_3 consumption (several T)
Ww-p+_T	wet weight, heat production (several T)
Ww-pX	wet weight, food energy consumption
Wd-Wa	dry weight, ash-free dry weight
Wd-Ww	dry weight, wet weight
Wd-JC_T	wet weight, CO_2 production (several T)
Wd-JO	dry weight, O_2 consumption
Wd-JO_f	dry weight, O_2 consumption (several f)
Wd-JO_T	dry weight, O_2 consumption (several T)
Wd-JN	dry weight, NH_3 production
Wd-JN_T	dry weight, NH_3 production (several T)
WC-JX	carbon weight,food consumption
JX-ap	ingestion rate, age at p
JX-am	ingestion rate, age at death
JX-Vp	ingestion rate, volume at p
JX-Vi	ingestion rate, volume at i
JX-Ri	ingestion rate, reproduction rate at i
JX-JO_L	ingestion rate, O_2 consumption (several L)
pX-pG	ingestion power, growth power
T-ah	temperature, age at h
T-ab	temperature, age at b
T-aj	temperature, age at j
T-ap	temperature, age at p
T-am	temperature, age at death
T-t	temperature, time between stages
T-astar	temperature, instar duration
T-dL	temperature, change in length
T-dWw	temperature, change in wet weight
T-dWw_f	temperature, change in wet weight (several f)
T-R	temperature, reproduction rate
T-N	temperature, number of eggs
T-JX	temperature, food consumption
T-JC	temperature, CO_2 production
T-JO	temperature, O_2 consumption
T-JO_f	temperature, O_2 consumption (several f)
T-JO_W	temperature, O_2 consumption (several W)
T-JO_fW	temperature, O_2 consumption (several f, W)
T-JN	temperature, NH_3 production